Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces

Class Test Two Solutions **26 April 2016**

Time : 18h00 – 19h30 **Total marks: 40**

1. (a) The eventually constant ones.
   (b) Yes
   (c) $([0,1], d_E)$ [3]

2. Suppose that A is a complete subset of the metric space (X,d). Take (a_n) a sequence in A such that $a_n \to x$ in (X,d). Since (a_n) converges, it is a Cauchy sequence in X and hence in A. By completeness of A, $a_n \to a$ for some $a \in A$. By uniqueness of limits, $x = a$ and so $x \in A$. [5]

3. (a) Suppose that $f : (X, d_X) \to (Y, d_Y)$ is Lipschitz-continuous, with Lipschitz constant k. Take (x_n) a sequence in X such that $x_n \to x$ in X.
   Now $0 \le d_Y(f(x_n), f(x)) \le k d_X(x_n, x)$ for all $n \in \mathbb{N}$.
   As $n \to \infty$, $d_X(x_n, x) \to 0$, so taking limits above gives $d_Y(f(x_n), f(x)) \to 0$.
   So $f(x_n) \to f(x)$, as required.

   (b) There are many possibilities, for instance:
   For any open set G of (Y, d_Y), $f^{-1}(G)$ is an open subset of (X, d_X).
   For any closed set K of (Y, d_Y), $f^{-1}(K)$ is an closed subset of (X, d_X).
   For any subset A of X, $f(\,\overline{A}\,) \subseteq \overline{f(A)}$.

   (c) No, f is not uniformly continuous.
   The sequences $(x_n) = (n)$ and $(y_n) = (n + \frac{1}{n})$ are neighbours in $([0,\infty), d_E)$, because $\left|n - (n + \frac{1}{n})\right| = \frac{1}{n} \to 0$ as $n \to \infty$.
   However, (x_n) and (y_n) are not neighbours in $([0,\infty), \rho)$, because $\left|n^2 - (n + \frac{1}{n})^2\right| = 2 + \frac{1}{n^2} \to 2$, not 0.

   (d) We use the Euclidean metric on the reals. The function $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ is a homeomorphism, since it is a bijection, it is continuous and its inverse, tan, is also continuous. However, $\mathbb{R}$ has infinite diameter, whereas $(-\frac{\pi}{2}, \frac{\pi}{2})$ has diameter π.

[4+4+4+3=15]

4. (a) For $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $m, n \geq N$ implies $d(x_n, x_m) < \epsilon$. Since $n_k \geq k$ and $n_\ell \geq \ell$, we have $k, \ell \geq N$ implies $d(x_{n_k}, x_{n_\ell}) < \epsilon$.

   (b) A is a complete subset of X.

   (c) A Cauchy sequence with a convergent subsequence converges itself, to the same limit as the subsequence.

   (d) Use $(\mathbb{R}, d_E)$ and $A_x = \{x\}$ for all $x \in (0, 1)$.

   (e) Use d_E on $A = \{\frac{1}{x} : x \in \mathbb{N}^+\}$ with $A_x = \{\frac{1}{x}\}$ for all $x \in \mathbb{N}^+$. Each of these singletons is open and complete, but their union A is not, because $(\frac{1}{n+1})$ is a Cauchy sequence in A that does not converge in A. [2+3+2+2+2=11]

5. (a) i. $\emptyset \in \mathcal{T}$ because $8 \notin \emptyset$ and $\mathbb{R} \in \mathcal{T}$ by definition.
If $8 \notin A$, then, for any $B \subseteq \mathbb{R}$, we also have $8 \notin A \cap B$, so $\mathcal{T}$ is closed under binary intersections.
Consider a collection $\{A_i : i \in I\}$ of members of $\mathcal{T}$.
If $8 \notin A_i$ for all $i \in I$, then $8 \notin \bigcup\{A_i : i \in I\}$.
If $A_j = \mathbb{R}$ for some $j \in I$, then $\bigcup\{A_i : i \in I\} = \mathbb{R}$.
So $\mathcal{T}$ is closed under arbitrary unions.

   ii. This space does not satisfy the Hausdorff property. It is not possible to find disjoint members of $\mathcal{T}$ separating 7 and 8, for example, because the only member of $\mathcal{T}$ containing 8 is $\mathbb{R}$.

   (b) The condition "$w_f(x) = 0$" holds if and only if f is continuous at x.

($\Longrightarrow$) Let $\epsilon > 0$ be given.
There exists G an open subset of X containing x with diam $f(G) < \epsilon$.
There exists $\delta > 0$ such that $B(x; \delta) \subseteq G$.
For $t \in B(x; \delta)$, we have $t \in G$ and so $f(t)$ and $f(x)$ are both in $f(G)$.
Then $d_Y(f(x), f(t)) \leq$ diam $f(G) < \epsilon$.
So, for any $t \in X$, $d_X(x, t) < \delta \Longrightarrow d_Y(f(x), f(t)) < \epsilon$.

($\Longleftarrow$) Let $\epsilon > 0$ be given.
There exists $\delta > 0$ such that, for any $t \in X$, $d_X(x, t) < \delta \Longrightarrow d_Y(f(x), f(t)) < \frac{\epsilon}{3}$.
This says, if $t \in B(x; \delta)$, then $f(t) \in B(f(x); \frac{\epsilon}{3})$.
Let $G = B(x; \delta)$. Then G is an open subset of X containing x and diam $f(G) \leq$ diam $B(f(x); \frac{\epsilon}{3}) \leq \frac{2\epsilon}{3} < \epsilon$.

[4+2=6]